

Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study

TEAM ID	SWTID-2026-6798
PROJECT TITLE	Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study
MAX MARKS	5 MARKS

CHAPTER – VII

FUNCTIONAL AND PERFORMANCE TESTING

7.1 – Performance Testing & Functional Testing

Testing ensures that the dashboard works correctly, provides accurate insights, and performs efficiently with housing datasets.

1. Functional Testing:

Functional testing verifies whether all features of the dashboard are working as expected.

Data Accuracy Testing:

Verify that sale prices, totals, and averages are calculated correctly.

Cross-check values with source data (CSV/Excel/database).

Filter Functionality Testing:

Check if filters (Year, Location, Price Range, Bedrooms) work properly

Ensure filtered results update all charts dynamically.

Chart Validation;

Confirm bar charts, line graphs, and maps display correct information.

Validate trend lines and calculated fields.

Data Connection Testing:

Ensure dataset is properly connected to Tableau.

Verify automatic refresh (if enabled).

User Interaction Testing:

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Test dashboard responsiveness when users select filters or parameters.

Ensure navigation between sheets works correctly.

Outcome: The dashboard functions correctly and provides accurate insights.

2. Performance Testing:

Performance testing ensures the dashboard loads quickly and works efficiently even with large datasets

Load Time Testing

Measure dashboard loading time.

Optimize if loading exceeds acceptable limits.

Data Volume Testing

Test performance with large housing datasets.

Check for slow rendering or lag

Query Optimization

Reduce Unnecessary calculated fields.

Optimize data source and remove unused columns.

Resource Utilization

Monitor CPU and RAM usage while running Tableau.

Ensure system meets minimum requirements.

Stress Testing

Test multiple filters and interactions simultaneously.

Ensure dashboard remains stable.

Outcome:

The dashboard performs efficiently without delays or crashes.